$$\hat{\beta}_1 \sim \text{norm}(\beta_1, \sigma^2/s_x^2)$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$s_x = \left(\sum_{i=1}^{n} (X_i - \bar{X})^2 \right)^{1/2}$$

$$T_{n-2} \sim s_x \frac{\hat{\beta}_1 - \beta_1^*}{\sigma'}$$

$$\sigma' = \left(\frac{s^2}{n-2} \right)^{1/2}$$

$$\frac{\hat{\beta}_1 - \beta_1^*}{\sigma' / s_x} \sim t_{n-2}$$

$$s^2 = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$